

Supply and load

A GCell has a routing budget, capacity

The idea

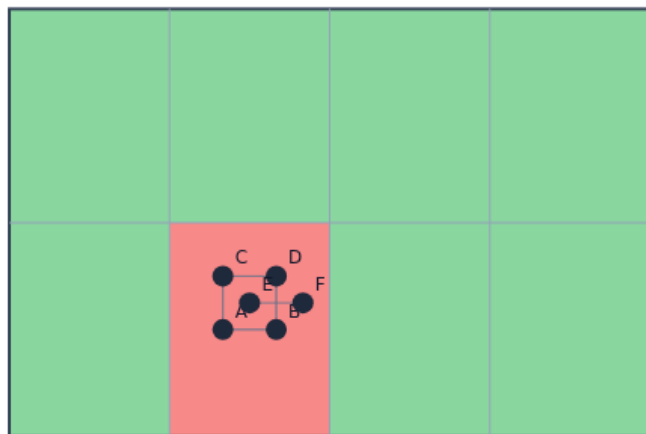
- Surplus equals demand minus capacity
- Positive surplus means overflow
- List oversubscribed tiles before you trust a heat map
- Capacity can later become anisotropic edge capacities

Capacity budget

CAPACITY VS DEMAND

Capacity budget

Step 1 / 5 · cap · module01-03-capacity-demand



Explanation

Each GCell has capacity 2.0 on the toy instance.

- Scalar Cap for goldens

Cap=2

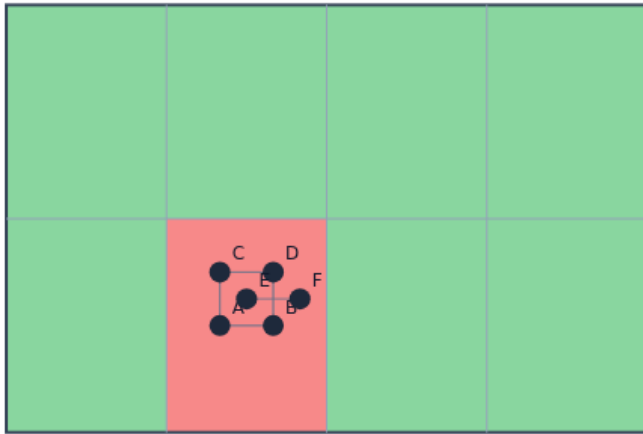
Capacity budget

Demand arrives

CAPACITY VS DEMAND

Demand arrives

Step 2 / 5 · demand · module01-03-capacity-demand



Explanation

Estimators deposit demand into tiles; $\text{surplus} = \text{demand} - \text{Cap.}$

- Positive surplus = overflow

Compare per tile

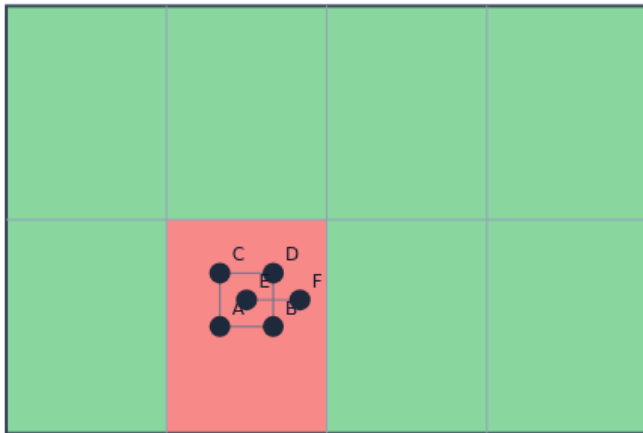
Demand arrives

Oversubscribed tiles

CAPACITY VS DEMAND

Oversubscribed tiles

Step 3 / 5 · flag · module01-03-capacity-demand



Explanation

Flag every tile with demand > Cap before trusting a heat map.

- List hot tiles

count ≥ 1 on seed

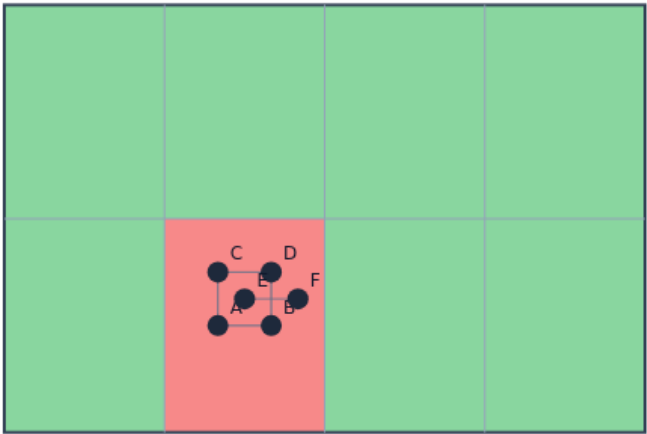
Oversubscribed tiles

Lower Cap

CAPACITY VS DEMAND

Lower Cap

Step 4 / 5 · toggle · module01-03-capacity-demand



Explanation

At Cap=1 more tiles fail—capacity is part of the contract.

- Document units

Cap is a knob

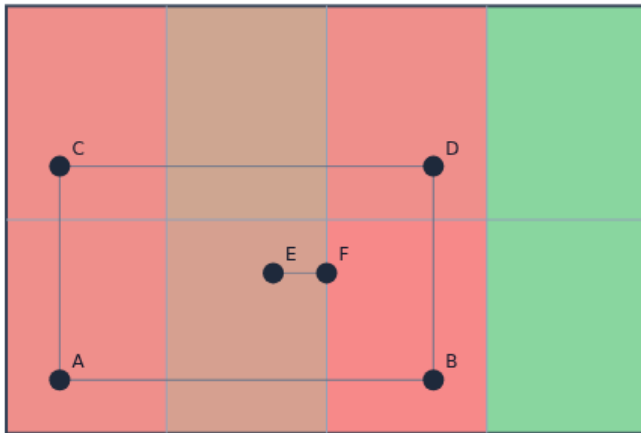
Lower Cap

Spread helps

CAPACITY VS DEMAND

Spread helps

Step 5 / 5 · spread · module01-03-capacity-demand



Explanation

Moving cells outward reduces how many tiles exceed Cap.

- Placement feeds congestion

Toward quieter tiles

Spread helps

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Capacity vs demand

INTERACTIVE TOOL

Capacity vs demand

Compare per-GCell demand to routing capacity and spot oversubscribed tiles.

Your job: Cap=2. Oversubscribed tiles have demand>2. Spread out until overflow count drops.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Seed has overflow

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

StartShow hintCheckNext challengeStop checkingIdle

A B C D E F←→↑↓Reset to starterReveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)
HPWL: 7.0
overflow total: 5.00 · max: 5.00 · count: 1
capacity: 2
view: your placement

Browser lab starter

Implement track

- Given a demand matrix from RUDY (or a hand-filled stub)
- Print both for capacity equals two and capacity equals one
- Keep the API ready for the overflow-metrics lab

Pitfalls

- Comparing demand to capacity without documenting units
- Treating zero demand as “healthy” while ignoring that neighboring tiles may be hot
- Changing capacity mid-challenge without resetting the starter placement

Your turn

- Complete Track A or B
- Next: RUDY, the classic demand estimator that fills those tiles from net bounding boxes